

Environment Variables Reference

This document lists every environment variable used across all three apps and their containers.

For production deployment, all variables are set in `.env.pg` (PostgreSQL stack) or `.env` (MySQL stack). See [DOCKER_SETUP.md](#) for how to create these files.

Backend Variables

Set in the `backend` and `migrate` services in the compose file.

Variable	Required	Default	Description
DATABASE_URL	Yes	—	Full database connection string. Prefix determines adapter: postgresql://... or mysql://...
SHADOW_DATABASE_URL	MySQL only	—	Shadow DB for Prisma migrations with MariaDB. Not needed for PostgreSQL.
JWT_SECRET	Yes	—	Secret key for signing/verifying JWT tokens. Must be changed from default. Generate with <code>openssl rand -hex 32</code> .
NODE_ENV	Yes	—	production or development. Affects logging and error output.

DATABASE_URL formats:

PostgreSQL:

```
postgresql://<user>:<password>@<host>:<port>/<database>
postgresql://pollinator:pollinatorpw@postgres:5432/pollinator
```

MySQL/MariaDB:

```
mysql://<user>:<password>@<host>:<port>/<database>
mysql://pollinator:pollinatorpw@mysql:3306/pollinator
```

The hostname (`postgres` or `mysql`) is the Docker service name. Do not use `localhost` in Docker — containers communicate by service name on the shared network.

Port: The backend listens on port 4000. This is hardcoded in [backend/src/index.ts](#) and is not configurable via environment variable. The port is internal to Docker and not exposed to the host.

Frontend Variables

Set in the `frontend` service in the compose file or in `frontend/.env.local` for local development outside Docker.

Variable	Required	Default	Description
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NEXT_PUBLIC_API_URL	Yes	""	Base URL for API calls. Must be empty string in production so Nginx proxies /api/... correctly. Set to http://localhost:4000 for direct-to-backend local dev (outside Docker).
NODE_ENV	Yes	—	production or development.

Why NEXT_PUBLIC_API_URL is empty in production: In the Docker stack, the frontend makes requests to /api/... (a relative path). Nginx intercepts these and proxies them to backend:4000. If NEXT_PUBLIC_API_URL is set to a full URL, requests bypass Nginx and will fail if the backend isn't directly accessible.

Development exception: When running the portal outside Docker (e.g., npm run dev on your machine while the backend runs separately), set NEXT_PUBLIC_API_URL=http://localhost:4000 to reach the backend directly.

Portal Variables

Same pattern as frontend.

Variable	Required	Default	Description
NEXT_PUBLIC_API_URL	Yes	""	Base URL for API calls. Empty in production (Nginx proxy on port 3001). Set to http://localhost:4000 for direct local dev.
NODE_ENV	Yes	—	production or development.

Database Container Variables

PostgreSQL

Set via env_file: .env.pg in docker-compose.prod.pg.yml.

Variable	Required	Description
POSTGRES_DB	Yes	Database name. Default in example: pollinator
POSTGRES_USER	Yes	Database user. Default in example: pollinator
POSTGRES_PASSWORD	Yes	Database password. Change from default.

These three must be consistent with the DATABASE_URL in the same .env.pg file:

```
DATABASE_URL=postgresql://${POSTGRES_USER}:${POSTGRES_PASSWORD}@postgres:5432/${POSTGRES_DB}
```

MySQL/MariaDB (development)

Hardcoded in [docker-compose.dev.yml](#) — not configurable without editing the compose file.

Variable	Value (hardcoded)	Description
MYSQL_ROOT_PASSWORD	rootpw	MySQL root password (dev only)
MYSQL_DATABASE	pollinator	Database name

MYSQL_USER	pollinator	App user
MYSQL_PASSWORD	pollinatorpw	App user password

For production MySQL, use `docker-compose.yml` (not `.dev.yml`) and provide these via an env file.

Environment Files

File	Used by	Notes
.env.pg.example	PostgreSQL stack	Template — copy to <code>.env.pg</code> and fill in secrets. Do not commit <code>.env.pg</code>.
<code>.env</code>	MySQL production stack	Not included — create from <code>docker-compose.yml</code> variable references
<code>frontend/.env.local</code>	Frontend dev (outside Docker)	Not committed. Set <code>NEXT_PUBLIC_API_URL=http://localhost:4000</code>
<code>portal/.env.local</code>	Portal dev (outside Docker)	Same as frontend

Variable Matrix

Variable	Backend	Frontend	Portal	MySQL DB	PostgreSQL DB
DATABASE_URL	✓				
SHADOW_DATABASE_URL	MySQL only				
JWT_SECRET	✓				
NODE_ENV	✓	✓	✓		
NEXT_PUBLIC_API_URL		✓	✓		
POSTGRES_DB/USER/PASSWORD					✓
MYSQL_ROOT_PASSWORD/DATABASE/USER/PASSWORD				✓	